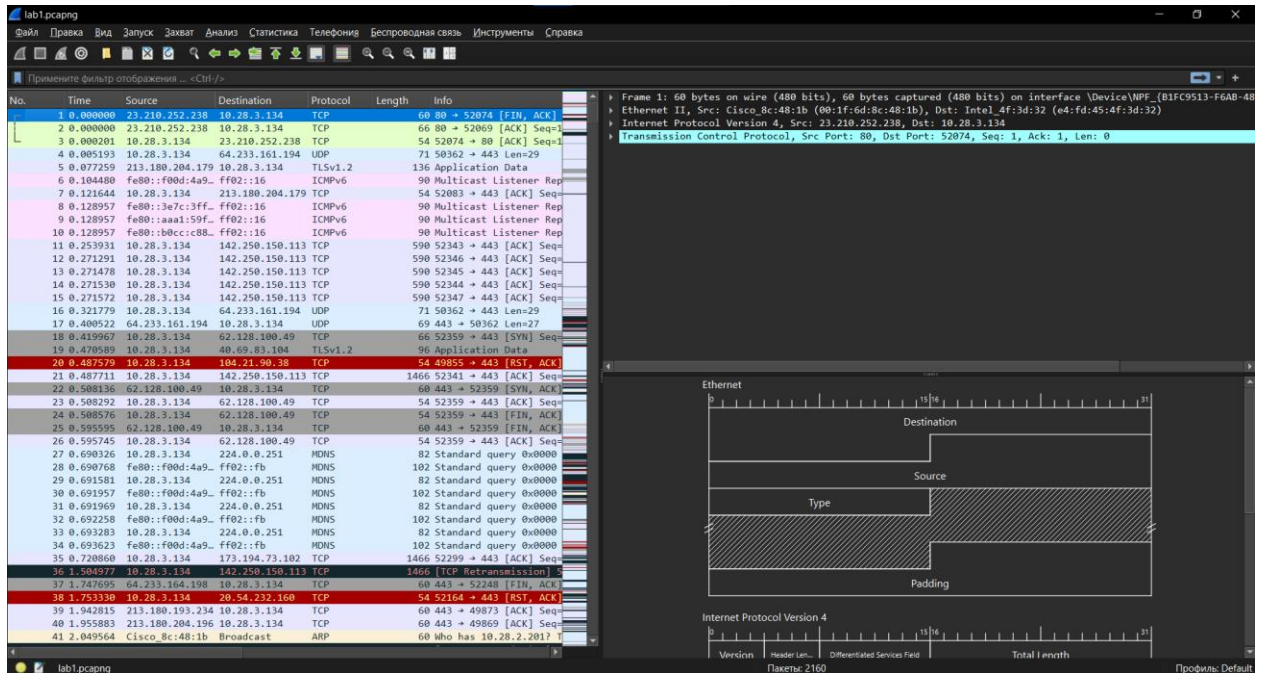


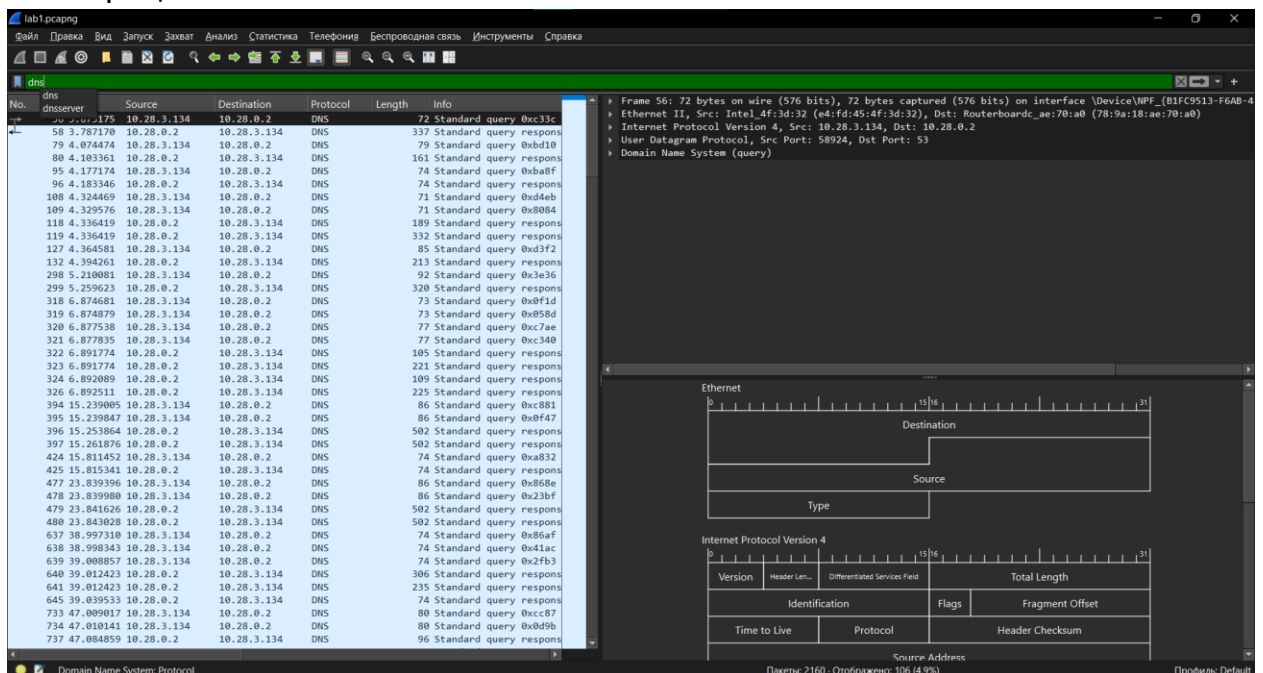
Отчёт по лабораторной работе №1 по Сетям ЭВМ

~Wireshark

1. Захват трафика



2. Фильтрация по DNS



3. Захват 10 пакетов

Wireshark capture of 10 packets. The packet list shows 10 packets, with packet 10 selected. The packet details pane shows the structure of an Ethernet II frame, an Internet Protocol Version 4 header, and a Transmission Control Protocol (TCP) segment. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	23.210.252.238	10.28.3.134	TCP	60	80 → 52074 [FIN, ACK] Seq=1
2	0.000000	23.210.252.238	10.28.3.134	TCP	60	80 → 52074 [ACK] Seq=1 Ack=1
3	0.000201	10.28.3.134	23.210.252.238	TCP	54	52074 → 80 [ACK] Seq=1 Ack=2
4	0.005193	10.28.3.134	64.233.161.194	UDP	71	50362 → 443 Len=29
5	0.077259	213.180.204.179	10.28.3.134	TLSv1.2	136	Application Data
6	0.104480	fe80::f0dd:4a9...	ff02::16	ICMPv6	90	Multicast Listener Report Me
7	0.121644	10.28.3.134	213.180.204.179	TCP	54	52089 → 443 [ACK] Seq=1 Ack=
8	0.128957	fe80::3e7c:3ff...	ff02::16	ICMPv6	90	Multicast Listener Report Me
9	0.128957	fe80::aaa1:59f...	ff02::16	ICMPv6	90	Multicast Listener Report Me
10	0.128957	fe80::b0cc:c88...	ff02::16	ICMPv6	90	Multicast Listener Report Me

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{B1FC9513-F6AB-48...} Ethernet II, Src: Cisco 8c:48:1b (00:1f:6d:8c:48:1b), Dst: Intel 4f:3d:32 (e4:fd:45:4f:3d:32) Internet Protocol Version 4, Src: 23.210.252.238, Dst: 10.28.3.134 Transmission Control Protocol, Src Port: 80, Dst Port: 52074, Seq: 1, Ack: 1, Len: 0

4. Захват 50 пакетов

Wireshark capture of 50 packets. The packet list shows 50 packets, with packet 50 selected. The packet details pane shows the structure of an Ethernet II frame, an Internet Protocol Version 4 header, and a Transmission Control Protocol (TCP) segment. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
10	4.417950	192.168.45.228	149.154.167.50	SSL	303	Continuation Data
11	4.747516	149.154.167.50	192.168.45.228	SSL	1294	Continuation Data
12	4.747516	149.154.167.50	192.168.45.228	SSL	1294	Continuation Data
13	4.747516	149.154.167.50	192.168.45.228	SSL	450	Continuation Data
14	4.747775	192.168.45.228	149.154.167.50	TCP	54	63212 → 443 [ACK] Seq=
15	4.749335	192.168.45.228	149.154.167.50	SSL	271	Continuation Data
16	4.950284	149.154.167.50	192.168.45.228	SSL	175	Continuation Data
17	4.999515	192.168.45.228	149.154.167.50	TCP	54	63212 → 443 [ACK] Seq=
18	5.361008	213.180.204.179	192.168.45.228	TLSv1.2	136	Application Data
19	5.417008	192.168.45.228	213.180.204.179	TCP	54	63252 → 443 [ACK] Seq=
20	5.800404	192.168.45.228	172.67.152.110	TCP	1454	[TCP Retransmission] 5
21	6.683393	192.168.45.228	172.67.152.110	TCP	1454	[TCP Retransmission] 5
22	12.683346	192.168.45.228	172.67.152.110	TCP	55	53107 → 443 [ACK] Seq=
23	13.416731	192.168.45.228	142.250.150.194	TCP	55	52504 → 443 [ACK] Seq=
24	13.552226	192.168.45.228	192.168.45.178	DNS	86	Standard query 0x8bb6
25	13.553402	192.168.45.228	192.168.45.178	DNS	86	Standard query 0x8bb6
26	13.556509	192.168.45.228	192.168.45.178	DNS	75	Standard query 0x1d41
27	13.557427	192.168.45.228	192.168.45.178	DNS	75	Standard query 0x3e11
28	13.558308	142.250.150.194	192.168.45.228	TCP	66	443 → 52504 [ACK] Seq=
29	13.583000	192.168.45.178	192.168.45.228	DNS	102	Standard query respons
30	13.583000	192.168.45.178	192.168.45.228	DNS	136	Standard query respons
31	13.586802	192.168.45.228	209.85.233.95	QUIC	1292	Initial, DCID=3ae6b88f
32	13.587144	192.168.45.228	209.85.233.95	QUIC	1292	Initial, DCID=3ae6b88f
33	13.588041	192.168.45.228	209.85.233.95	TCP	66	53263 → 443 [SYN] Seq=
34	13.588527	192.168.45.178	192.168.45.228	DNS	171	Standard query respons
35	13.588527	192.168.45.178	192.168.45.228	DNS	125	Standard query respons
36	13.589925	192.168.45.228	209.85.233.138	TCP	66	53264 → 443 [SYN] Seq=
37	13.690231	209.85.233.95	192.168.45.228	TCP	66	443 → 53263 [SYN, ACK]
38	13.690231	209.85.233.138	192.168.45.228	TCP	66	443 → 53264 [SYN, ACK]
39	13.690639	192.168.45.228	209.85.233.95	TCP	54	53263 → 443 [ACK] Seq=
40	13.690783	192.168.45.228	209.85.233.138	TCP	54	53264 → 443 [ACK] Seq=
41	13.691977	192.168.45.228	209.85.233.138	TLSv1.2	1875	Client Hello (SNI=play
42	13.693349	192.168.45.228	209.85.233.95	TLSv1.3	2352	Client Hello (SNI=uaa
43	13.885027	192.168.45.228	209.85.233.138	TCP	66	53265 → 443 [SYN] Seq=
44	13.888549	209.85.233.95	192.168.45.228	TCP	54	443 → 53263 [ACK] Seq=
45	13.888549	209.85.233.95	192.168.45.228	TCP	54	443 → 53263 [ACK] Seq=
46	13.888549	209.85.233.95	192.168.45.228	TLSv1.3	1454	Server Hello, Change C
47	13.888549	209.85.233.95	192.168.45.228	TLSv1.3	129	Application Data
48	13.888841	192.168.45.228	209.85.233.95	TCP	54	53263 → 443 [ACK] Seq=
49	13.890164	192.168.45.228	209.85.233.95	TLSv1.3	128	Change Cipher Spec, Ap
50	13.890873	192.168.45.228	209.85.233.95	QUIC	1292	Initial, DCID=3ae6b88f

Frame 1: 1454 bytes on wire (11632 bits), 1454 bytes captured (11632 bits) on interface \Device\NPF_{B1FC9513-F6AB-48...} Ethernet II, Src: Intel 4f:3d:32 (e4:fd:45:4f:3d:32), Dst: 5a:d4:92:5f:ec:65 (5a:d4:92:5f:ec:65) Internet Protocol Version 4, Src: 192.168.45.228, Dst: 172.67.152.110 Transmission Control Protocol, Src Port: 53198, Dst Port: 443, Seq: 1, Ack: 1, Len: 1400

~Mikrotik

1. Изменение имени на mt-00 и установка режима выключения экрана при бездействии поль-ля в 60 минут

```
[admin@MikroTik] > /system identity set name="mt-00"
[admin@mt-00] > /system console screen set blank-interval=1m
[admin@mt-00] > /system console screen print
    line-count: 25
    blank-interval: 1min
[admin@mt-00] > /system console screen set blank-interval=60m
[admin@mt-00] > /system console screen print
    line-count: 25
    blank-interval: 60min
```

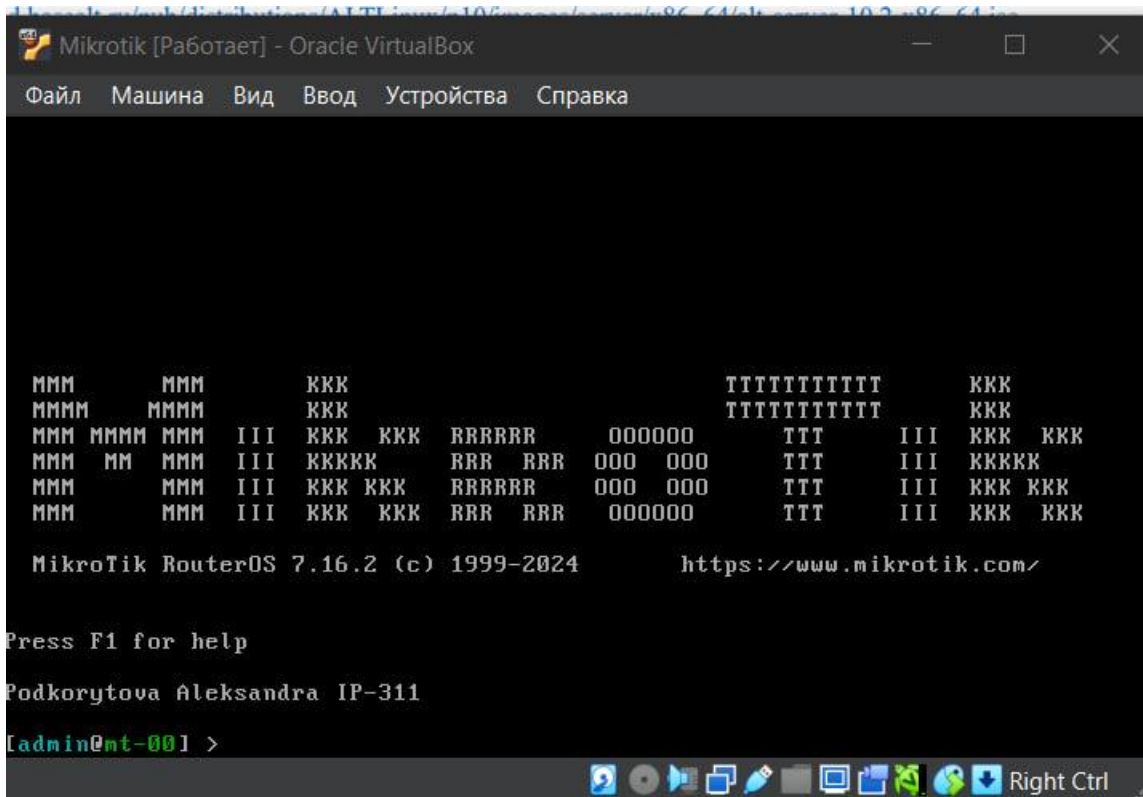
2. Установка информационного сообщения (вывод после входа в систему)

```
[admin@mt-00] > /system note set show-at-login=yes note="Podkorytova Aleksandra IP-311"
[admin@mt-00] > /system note print
    show-at-login: yes
    show-at-cli-login: no
    note: Podkorytova Aleksandra IP-311
[admin@mt-00] > _
```

3. Установка информационного сообщения (вывод до входа в систему)

```
[admin@mt-00] > /system note set show-at-cli-login=yes
[admin@mt-00] > /system note print
    show-at-login: yes
    show-at-cli-login: yes
    note: Podkorytova Aleksandra IP-311
```

Результат:



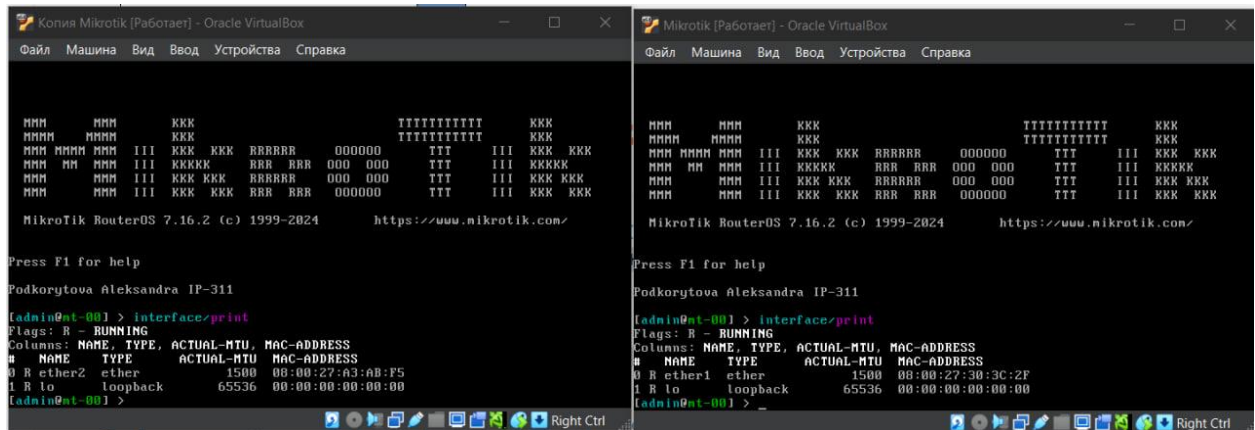
4. Вывод информации о пользовательских группах

```
[admin@mt-00] > /user print
Columns: NAME, GROUP, LAST-LOGGED-IN, INACTIVITY-POLICY
# NAME  GROUP  LAST-LOGGED-IN  INACTIVITY-POLICY
::: system default user
0 admin full  2025-02-12 07:20:01 none
[admin@mt-00] > /user group print
0 name="read" policy=local,telnet,ssh,reboot,read,test,winbox,password,web,
  sniff,sensitive,api,romon,rest-api,!ftp,!write,!policy
  skin=default

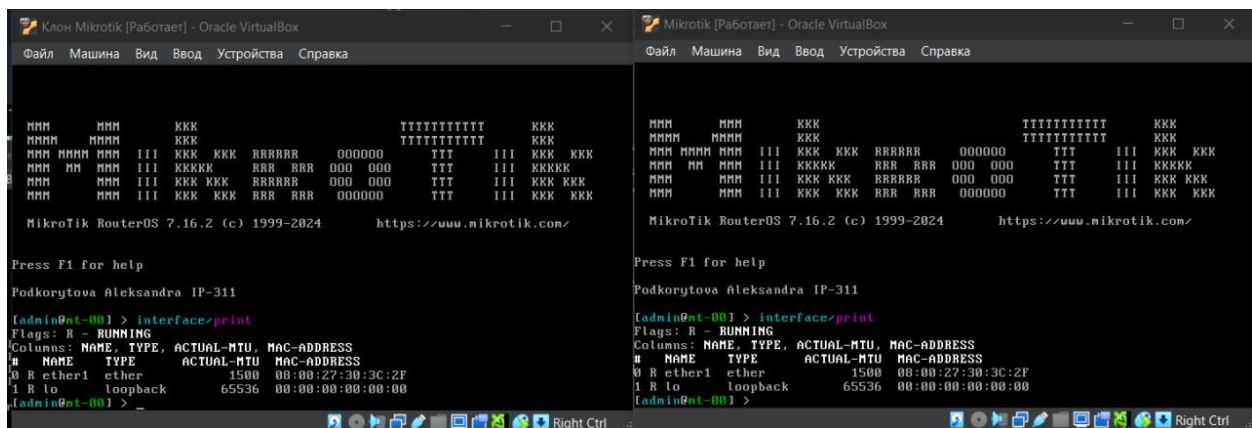
1 name="write" policy=local,telnet,ssh,reboot,read,write,test,winbox,password,
  web,sniff,sensitive,api,romon,rest-api,!ftp,!policy
  skin=default

2 name="full" policy=local,telnet,ssh,ftp,reboot,read,write,policy,test,winbox,
  password,web,sniff,sensitive,api,romon,rest-api
  skin=default
[admin@mt-00] >
```

5. Отличие копии VM от «родителя»



6. Отличие клона от «родителя»



~Базальт ОС

1. Регистрация от имени администратора и команда вывода всех записей network


```
Welcome to ALT Server 10.2 (Mendelevium)?

Hostname: host-basalt
IP: 10.0.2.15
host-basalt login: root
Password:
Last login: Tue Feb 11 20:28:38 +07 2025 on tty1
[root@host-basalt ~]# journalctl | grep -i network_
```

2. Вывод записей от имени поль-ля

```
libcairo-1.16.0      rpcbind-1.2.6
libcap-ng-0.8.2      rpm-4.13.0.1
libcroco-0.6.13      rsync-3.2.7
libcrypt             sash-3.4
libcurl-8.5.0        screen-4.8.0
libdeflate-1.10      sdparm-1.12
libelf-0.185         sed-4.8.0.7.b352
libfreetype-2.11.0   setproctitle-0.3.2
libgc-7.6.8          sg3_utils-1.47
libgcrypt20-1.9.3    shadow-utils-4.5
libgcrypt-gost20-1.8.5 shared-mime-info-2.2
libgdbm-1.8.3        slang-2.3.2
libgmp10-6.2.1       smartmontools-7.2
libgpg-error-1.42    stmpclean-0.3
libgraphite2-1.3.14  strace-6.7
libgsasl-1.8.0       su-0.60
libidn-1.37          sudo-1.9.15p1
libidn2-2.3.2        sysstat-12.7.4
libinih0-r53         systemd-249.17
libjpeg-turbo        tar-1.34.0.16.12d67f44
libksba-gost-1.3.6   tcl-8.6.11
liblcms2-2.12        tcpdump-4.9.3
libltdl-2.4.6        time-1.7
liblua5.3-5.3.5      tinycdb-0.78
liblz4-1.9.3         tzdata-2022g
liblz02              unzip-6.0
libmnl-1.0.4         usbutils-014
libmspack-0.6        util-linux-2.39.2
libnatspec-0.3.1     vim-common-9.0.2136
libnettle8-3.7.3     which-2.20
libnfnctlink-1.0.1.0.8.5087 xfsprogs-5.12.0
libntlm-1.5          xz-5.2.5
libopenjpeg2.0-2.5.0 zip-3.0
libp11-kit-0.24.1    zlib-1.2.13
libpaper-1.1.28      zstd-1.5.5
libpasswdqc-2.0.2
[user@host-basalt ~]# ls /usr/share/doc
```

3. Вывод всех записей network

```
Feb 19 14:27:46 host-basalt systemd[1]: network.service: Deactivated successfully.
Feb 19 14:27:46 host-basalt systemd[1]: Stopped Network Connectivity.
Feb 19 14:27:46 host-basalt systemd[1]: Stopped target Preparation for Network.
Feb 19 18:57:12 host-basalt systemd[1]: Reached target Preparation for Network.
Feb 19 18:57:12 host-basalt (udev-worker)[2169]: device-mapper: /lib/udev/rules.d/10-dm.rules:31 Only network interfaces can be renamed, ignoring NAME="mapper/control".
Feb 19 18:57:12 host-basalt kernel: e1000: Intel(R) PRO/1000 Network Driver
Feb 19 18:57:12 host-basalt (udev-worker)[2174]: device-mapper: /lib/udev/rules.d/10-dm.rules:31 Only network interfaces can be renamed, ignoring NAME="mapper/control".
Feb 19 18:57:12 host-basalt udevadm[2152]: systemd-udev-settle.service is deprecated. Please fix network.service not to pull it in.
Feb 19 18:57:13 host-basalt kernel: e1000 0000:00:03:0 eth0: Intel(R) PRO/1000 Network Connection
Feb 19 18:57:13 host-basalt systemd[1]: Starting Network Connectivity...
Feb 19 18:57:13 host-basalt network[2254]: Computing interface groups: .. 2 interfaces found
Feb 19 18:57:13 host-basalt network[2254]: Starting group 0/virtual (1 interfaces)
Feb 19 18:57:13 host-basalt network[2254]: Starting lo:
Feb 19 18:57:13 host-basalt network[2284]: 'lo' is already up
Feb 19 18:57:13 host-basalt network[2254]: SKIPPED
Feb 19 18:57:13 host-basalt network[2254]: Starting group 1/realphys (1 interfaces)
Feb 19 18:57:13 host-basalt network[2254]: Starting enp0s3:
Feb 19 18:57:13 host-basalt network[2305]: .
Feb 19 18:57:13 host-basalt network[2317]: .
Feb 19 18:57:13 host-basalt network[2319]: .
Feb 19 18:57:13 host-basalt network[2327]: dhcpd-10.0.6 starting
Feb 19 18:57:13 host-basalt network[2330]: enp0s3: waiting for carrier
Feb 19 18:57:15 host-basalt network[2330]: enp0s3: carrier acquired
Feb 19 18:57:17 host-basalt network[2330]: enp0s3: rebinding lease of 10.0.2.15
Feb 19 18:57:17 host-basalt network[2330]: enp0s3: leased 10.0.2.15 for 86400 seconds
Feb 19 18:57:17 host-basalt network[2330]: enp0s3: adding route to 10.0.2.0/24
Feb 19 18:57:17 host-basalt network[2330]: enp0s3: adding default route via 10.0.2.2
Feb 19 18:57:17 host-basalt network[2319]: .
Feb 19 18:57:17 host-basalt network[2317]: .
Feb 19 18:57:17 host-basalt network[2254]: OK
Feb 19 18:57:17 host-basalt network[2254]: Processing /etc/net/unlatab: empty.
Feb 19 18:57:17 host-basalt systemd[1]: Started Network Connectivity.
Feb 19 18:57:17 host-basalt systemd[1]: Reached target Network.
[root@host-basalt ~]# journalctl | grep -i networkSS
```

4. Вывод записей LIB

```
LIBnetlink-1.0.1.0.8.5087
LIBntlm-1.5
LIBopenjpeg2.0-2.5.0
LIBp11-kit-0.24.1
LIBpaper-1.1.28
LIBpasswdqc-2.0.2
LIBpcap-1.9.1
LIBpcre2-10.42
LIBpipeline-1.5.3
LIBpng-1.6.37
LIBpoppler130-23.08.0
LIBpsl-0.21.1
LIBqpdf28-10.4.0
LIBsas12-3-2.1.27
LIBseccomp-2.5.4
LIBsgutils-1.47
LIBshell-0.4.11
LIBssh2-1.11.0
LIBtasn1
LIBtcb-1.2
LIBtiff-4.4.0
LIBtirpc-1.3.3
LIBunistring2-0.9.10
LIBuring-2.2
LIBusb-1.0.26
LIButempter-1.2.1
LIBuerto-0.3.2
LIBX11-1.8.7
LIBxcb-1.14
LIBxml2-2.9.12
LIBxmlsec1-1.2.37
LIBxslt-1.1.34
LIBxxhash-0.8.0
LIBzio-1.06
LIBzstd-1.5.5
zLIB-1.2.13
[user@host-basalt ~]$ ls /usr/share/doc | grep "lib" | sed 's/lib/LIB/g'
```

5. Код скрипта в BASH (написан в vim)


```
if [ $# -eq 0 ]; then
    echo "no params"
    exit 1
fi

echo "params:"
index=1
for param in "$@"; do
    echo "param $index: $param"
    ((index++))
done_

-- INSERT --
```

6. Запуск скрипта

```

"script.sh" [New] 11L, 145B written
[root@host-basalt ~]# ./script.sh hello world 123
-bash: ./script.sh: Permission denied
[root@host-basalt ~]# script.sh par1 par2 par3
-bash: script.sh: command not found
[root@host-basalt ~]# chmod u+x script.sh
[root@host-basalt ~]# script.sh par1 par2 par3
-bash: script.sh: command not found
[root@host-basalt ~]# ls -l script.sh
-rwxr--r-- 1 root root 145 Feb 19 19:32 script.sh
[root@host-basalt ~]# .script.sh
-bash: .script.sh: command not found
[root@host-basalt ~]# ./script.sh
no params
[root@host-basalt ~]# script.sh par1 par2 par3
-bash: script.sh: command not found
[root@host-basalt ~]# /script.sh par1 par2 par3
-bash: /script.sh: No such file or directory
[root@host-basalt ~]# ./script.sh par1 par2 par3
params:
param 1: 3
[root@host-basalt ~]# _

```

7. После регистрации от имени администратора, используя потоковый редактор sed удалила в файле /etc/sudoers символы "# " (решетка и пробел) в начале строк содержащих подстроку SUDO_USERS. Добавила группу пользователей sudo и включила в неё локального пользователя.

```

Welcome to ALT Server 10.2 (Mendelevium)!

hostname: host-basalt
IP: 10.0.2.15
host-basalt login: root
Password:
Last login: Wed Feb 19 19:14:31 +07 2025 on tty1
[root@host-basalt ~]# sed -i ' /SUDO_USERS/s/^#//' /etc/sudoers
[root@host-basalt ~]# groupadd sudo
[root@host-basalt ~]# usermod -aG sudo sasha
usermod: user 'sasha' does not exist
[root@host-basalt ~]# usermod -aG sudo root
[root@host-basalt ~]#

```

~Astra Linux

1. На этом этапе произошла регистрация, настройка среды, а также создание рабочей директории.
 - 1) Создана директория project через mkdir
 - 2) В этой директории создан пустой текстовый документ через команду touch file.txt

- 3) Через `git add file.txt` проиндексировала файл
- 4) Через `git commit -m "Предполагаемый текст"` создан коммит моих действий
- 5) По умолчанию была создана ветка `master`

```
31a9ad539e0a0b86bcc6e0e1109c6f4ef3f75871) $ git log
commit 31a9ad539e0a0b86bcc6e0e1109c6f4ef3f75871
Author: Podkorytova Aleksandra IP-311 <sasha.podkoritova2018@yandex.ru>
Date:   Wed Feb 19 20:14:58 2025 +0700

    Add empty file.txt
```

2. Настройка BASH для отображения текущей ветки (код в nano)

```
GNU nano 2.7.4                                Файл: /home/user/.bashrc

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
fi
fi

__git_ps1(){
    local branch
    branch=$(git rev-parse --abbrev-ref HEAD 2>/dev/null)
    if [ $? -eq 0 ]; then
        printf "(%s)" "$branch"
    fi
}

export PS1='\u@\h:\w$(__git_ps1) \> '
```

Результат:

```
Astra Linux CE 2.12.46 (orel) host-astra tty1
host-astra login: user
Password:
Last login: Thu Feb 20 08:25:56 +07 2025 on tty1
You have new mail.
user@host-astra:~ $ cd project/
user@host-astra:~/project(--abbrev-ref
31a9ad539e0a0b86bcc6e0e1109c6f4ef3f75871) $
```

